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# Alabama Aerodynamics Experimental Research Operations: *Evaluation and Comparison of Propeller Designs for UAVs*

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## ABSTRACT

The Alabama Aerodynamics Experimental Research Operations (A.A.E.R.O.) group is a team dedicated to advancing drone propulsion technology through research and development. The objective of the A.A.E.R.O. group is to investigate the aerodynamic performance of four distinct propeller designs—conventional (planar), alular, toroidal, and forward-swept—to determine their impact on thrust generation and noise reduction. The central research question driving this study is: *How do novel propeller geometries compare to conventional designs in terms of aerodynamic efficiency and acoustic performance?* This research is motivated by the growing need for more efficient, quieter propulsion systems in unmanned aerial vehicles (UAVs). The methodology integrates propeller testing, simulations and in-house manufacturing for experimental validation. Preliminary findings suggest that both toroidal and forward-swept propellers may offer improvements in noise reduction while maintaining competitive thrust generation. Future research will explore aerodynamic optimization and assess real-world applications in UAV technology. This study lays the groundwork for further innovation in aerodynamics, with the potential to impact both commercial and defense industries.

## INTRODUCTION

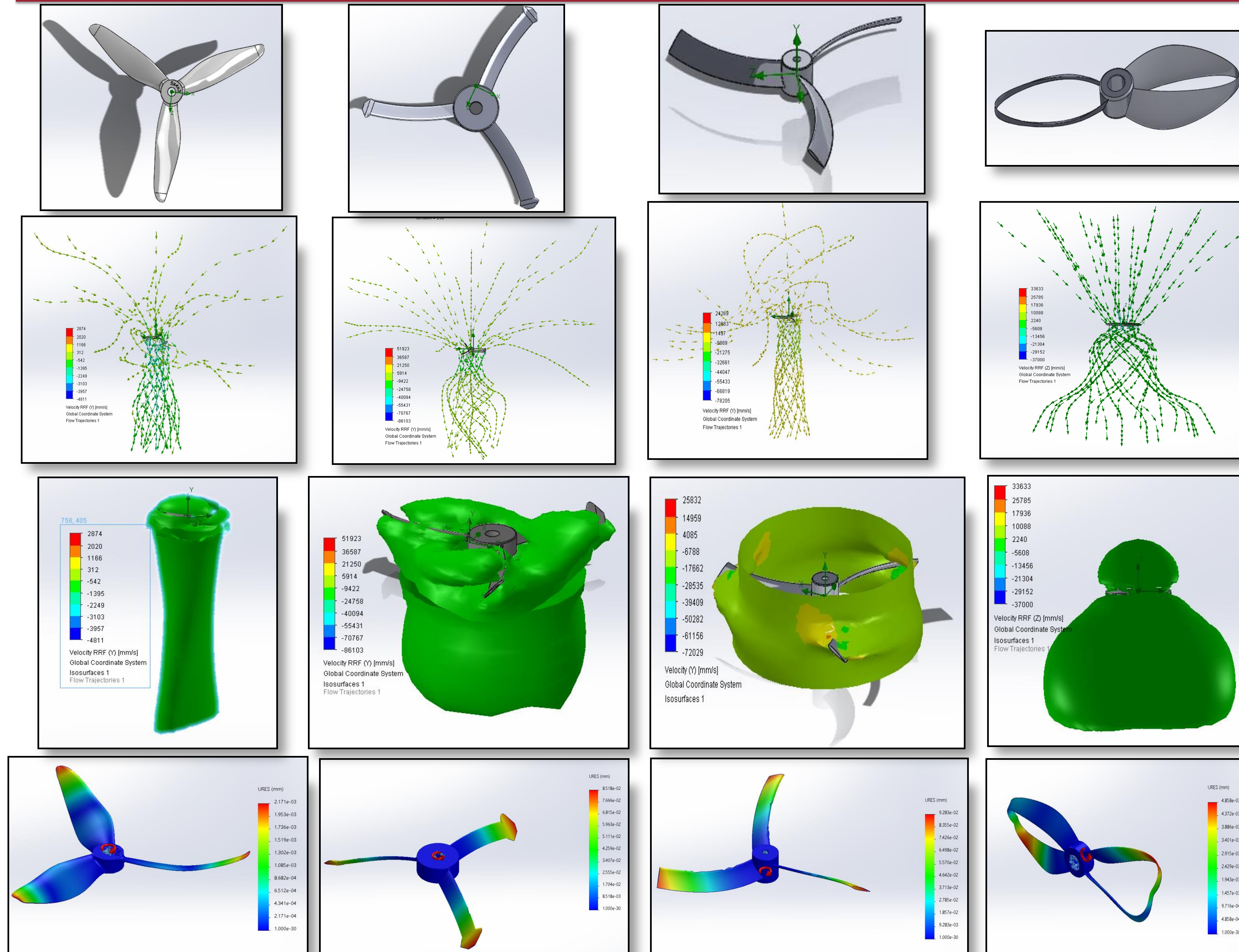
As the demand for more efficient and quieter propulsion systems in unmanned aerial vehicles (UAVs) continues to grow, innovative propeller designs have emerged as a key area of research. Traditional planar propellers, while effective in thrust generation, often contribute to significant noise pollution, which poses challenges for both commercial and military applications. To address this issue, A.A.E.R.O. aims to investigate alternative propeller geometries, with an initial focus on thrust production. By integrating computational simulations, in-house manufacturing, and experimental testing, the research seeks to optimize propeller performance for various aerial platforms. The findings will contribute to the ongoing advancements in UAV technology.

## METHODOLOGY

The methodology involves developing the four distinct propeller geometries using CAD software (SolidWorks and ANSYS Design Modeler) to assess aerodynamic efficiency:

- Computational fluid dynamics (CFD) simulations in SolidWorks replicate airflow behavior at 10,000 RPM.
- A computational grid calculates flow properties such as pressure, exit velocity, thrust, and Mach number (M). Thrust analysis quantifies net force, while pressure and velocity contours characterize airflow.
- Optimization efforts refine angle of attack and tip/surface designs to enhance thrust and minimize drag.
- Performance metrics are compared across designs, with results validated against theoretical models and existing research.

## RESULTS



The Finite Element Analysis (FEA) simulations reveal distinct displacement patterns across the propellers. The toroidal propeller exhibits the highest displacement ( $9.283 \times 10^{-2}$  mm) at its outer edges, indicating flexibility. The forward sweep ( $7.666 \times 10^{-2}$  mm) and default ( $5.818 \times 10^{-2}$  mm) propellers show moderate deformation, while the alular propeller ( $8.515 \times 10^{-3}$  mm) demonstrates the least displacement, suggesting superior rigidity. CFD analysis in SolidWorks reveals performance differences. The alular propeller shows the highest velocity ( $-162$  mm/s) and normal force ( $4.568$  N in Y), indicating strong thrust. The default propeller has the lowest velocity ( $-354$  mm/s) but a normal force of  $1.143$  N. The forward sweep (velocity  $-157$  mm/s, force  $4.170$  N) and toroidal (velocity  $-150$  mm/s, force  $1.390$  N in Y) exhibit balanced performance.

## CONCLUSIONS/SIGNIFICANCE

The experimental analyses of the four propellers reveal trade-offs between thrust efficiency and noise reduction across VTOL, helicopters, and conventional aircraft. Alular propellers generate the highest thrust, while toroidal designs significantly reduce noise but suffer from lower aerodynamic performance. Forward-sweep and alular configurations balance efficiency, making it ideal for high-speed UAVs. Toroidal propellers reduce blade-vortex interactions, meeting noise regulations but requiring optimization for practical VTOL use. In conventional aircraft, forward-sweep designs enhance fuel efficiency. Further refinements through computational optimization will enhance thrust, reduce drag, and improve noise performance for UAV propulsion.

## FUTURE DIRECTIONS

Future research will focus on refining the novel propeller designs to enhance efficiency, particularly for the toroidal propeller, while exploring innovative configurations like the V-rotor and varying blade numbers. Additionally, Fast Fourier Transform (FFT) analysis will be used to gain a better understanding of the sound produced by these propellers. By combining detailed simulations, 3D-printed prototypes, and experimental data from wind-tunnel testing, existing models will be improved. As an extension, new designs will be developed, advancing propeller technology for UAV and VTOL applications.

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